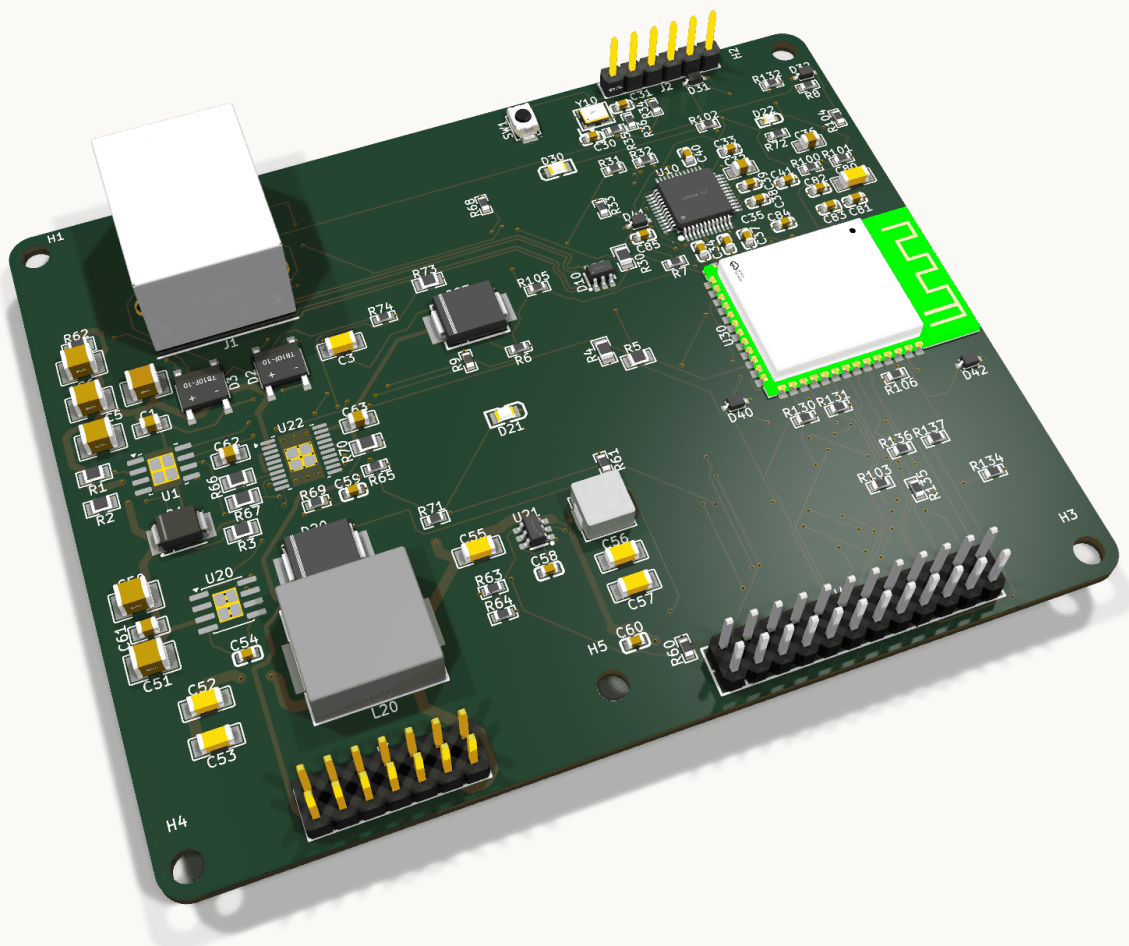


hwde

An AI PCB engineer for KiCad and JLCPCB

printed circuit boards, designed from a written brief

You describe the board you want in plain words. hwde researches the parts, picks a topology, draws the schematic, lays out and routes the board, runs the electrical and manufacturing checks, and hands you the files a factory needs to build it. A human engineer signs off at every stage.

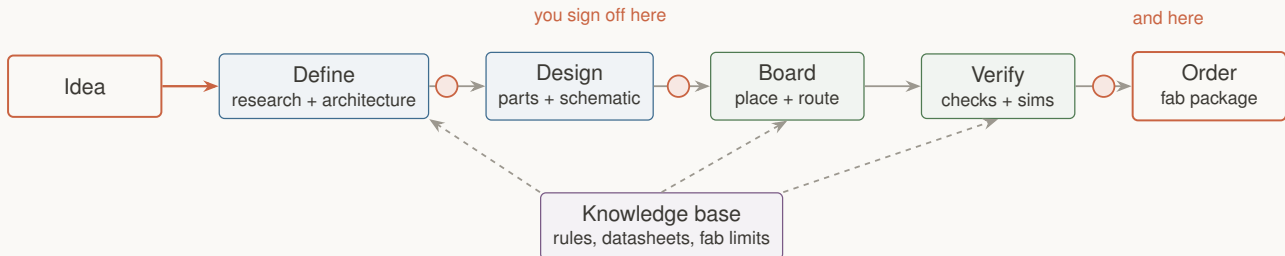


lumina-carrier — a 4-layer, 100 × 80 mm carrier board for a stage-lighting fixture. 116 footprints. Designed by hwde, ordered from JLCPCB and built.

How it works

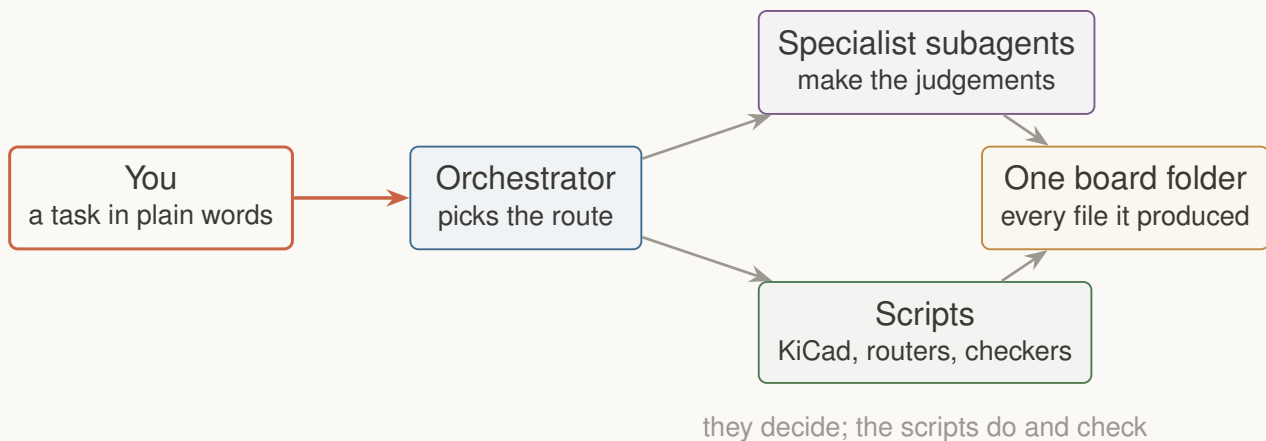
Three pictures. The first is the route a board takes, the second is who does which part of the work, the third is what happens at every handover.

The pipeline



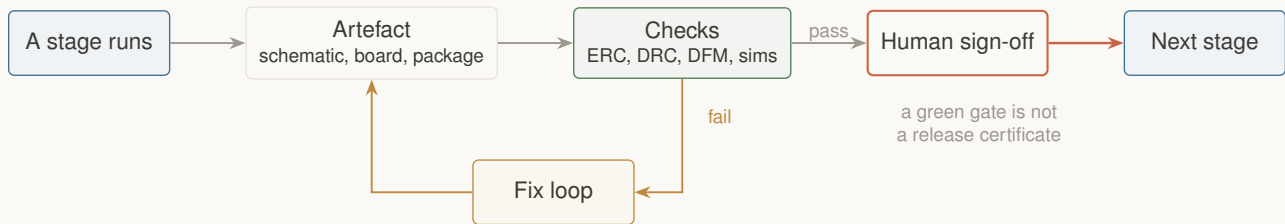
A run starts from a written brief and ends with a manufacturing package: Gerbers, a bill of materials and a pick-and-place file. Each middle stage produces one artefact, and reads the knowledge base of design rules, extracted datasheet numbers and the factory's real capabilities. The orange dots are where a person has to look at the work and agree before it continues.

Who does what



The split matters. Twenty-two specialist subagents make the judgement calls — choosing a topology, sizing a part, reviewing a schematic — and each has a written contract for what it may decide. Sixty-odd deterministic scripts do and check the mechanical work: they drive KiCad, run the routers, export the Gerbers. A script never has an opinion, an agent never edits a board file by hand, and everything either produces lands as a file in one folder per board.

The gate loop



Every stage ends at a gate, which runs the checks that apply: electrical rules, design rules, manufacturability, and circuit simulations against stated bounds. It either passes, or it hands the findings back to a fix loop that reworks the artefact and tries again — and a pass still needs a person's sign-off before the next stage starts. A green gate proves the checks that ran, not the checks it lists, so the reports are kept per check rather than collapsed into a verdict.

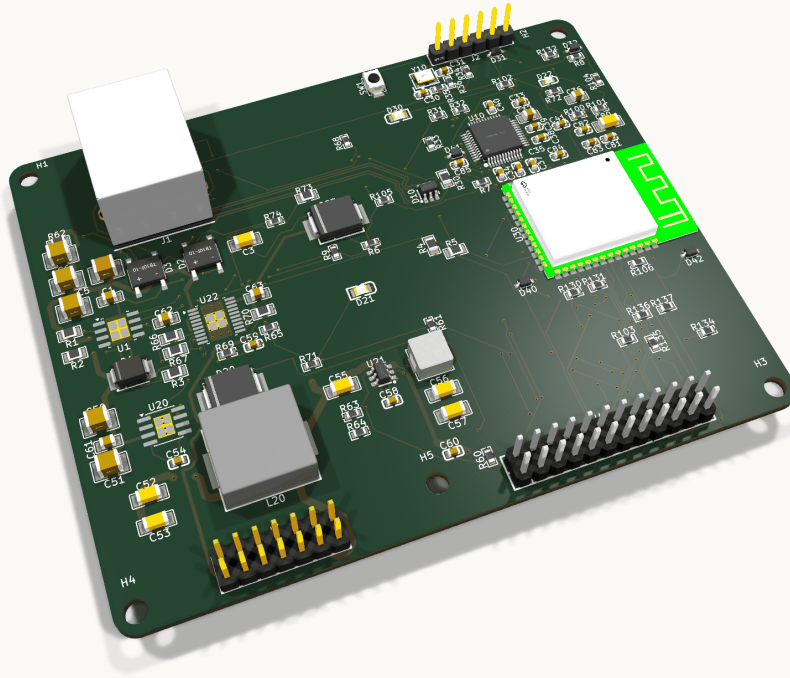
The boards

Fourteen boards have been taken far enough to have a laid-out, routed PCB. They run from a five-part linear regulator to a 200 W RF power stage. Every render below is made from the board file in the repository, by the script next to this document.

Board	What it is	Layers	Size (mm)	Stage reached
lumina-carrier	lighting carrier, PoE-powered	4	100 × 80	ordered, built
pd-trigger	USB-C PD bench trigger	2	48 × 30	ordered, built
lumina-par	RGBW PAR daughter board	4	100 × 80	verification
rf-de-20m	20 MHz Class E GaN stage	4	120 × 80	manufacturability
sbuck-5v3a	5 V / 3 A synchronous buck	4	50 × 40	package ready
usb-buck	USB device dev board	4	50 × 40	package ready
g0-sense	USB-C sensor node	2	35.8 × 28.3	package ready
stm32-blinky	minimal STM32 dev board	2	50 × 40	package ready
rf-term-150w	150 W RF dummy load head	2	26 × 20	manufacturability
bb-adc	single-channel ADC	2	54.8 × 34.9	manufacturability
bb-amp	sensor amplifier chain	2	48 × 28.3	manufacturability
bb-buck	bare buck converter	2	35 × 25	manufacturability
bb-ldo	bare linear regulator	2	34.7 × 34.7	manufacturability
bb-mcu	bare microcontroller board	2	34.8 × 22.3	manufacturability

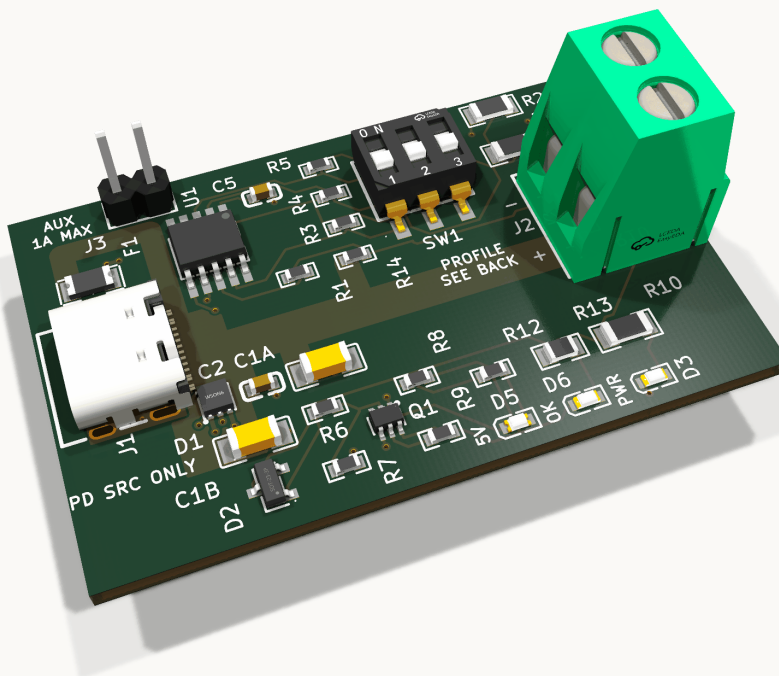
“Stage reached” is where the board stopped in the pipeline, not a statement that it is fit to release. Two boards say “ordered, built” because their own files record a JLCPCB order number that tracked through to shipped. The rest stopped with a checked manufacturing package in hand. Sizes are the bounding box of the board outline; not every outline is a rectangle.

Designed, ordered and built



lumina-carrier 4 layers 100 × 80 mm 116 footprints ordered from JLCPCB, tracked to shipped

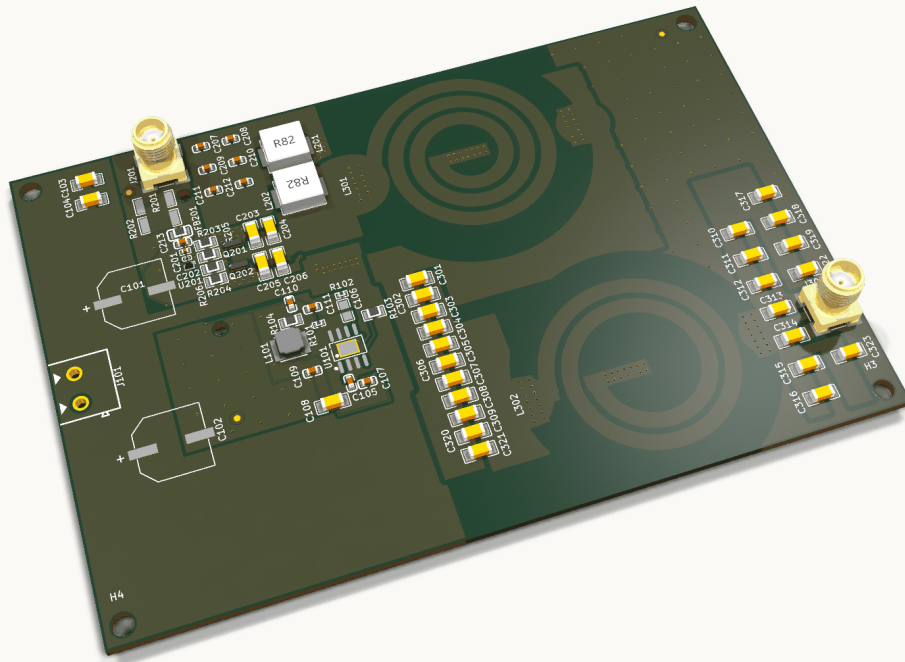
The universal carrier for a stage-lighting fixture. It takes power over Ethernet through a PD controller and a 100 V buck, and carries a frozen expansion connector that a fixture-specific daughter board plugs into. The biggest board here: 4 layers, 116 footprints, a controlled interface another board depends on.



pd-trigger 2 layers 48 × 30 mm 30 footprints ordered from JLCPCB, tracked to shipped

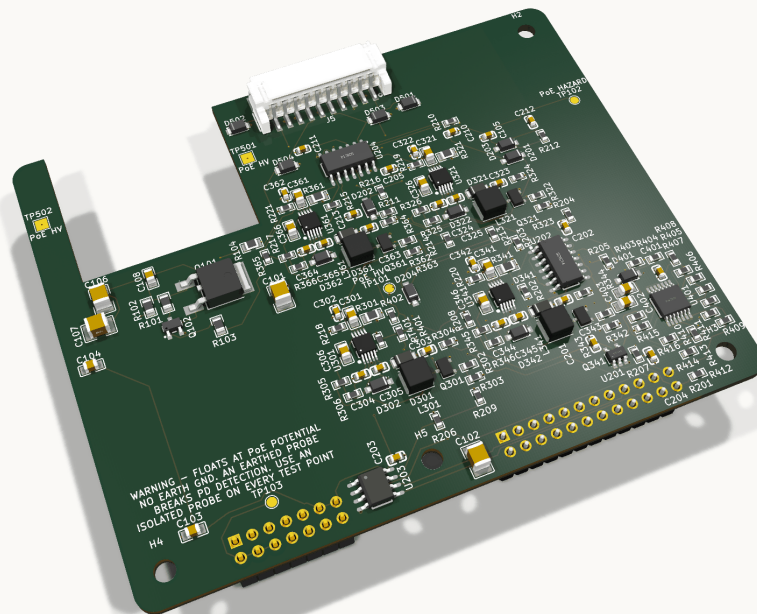
A bench tool. A USB-C Power Delivery sink controller negotiates with a charger for 5, 9, 12, 15 or 20 volts, you pick the profile on the board, and the negotiated rail comes out on a screw terminal rated for 5 A — up to 100 W. No conversion on the board; it is a pass-through trigger.

Harder boards



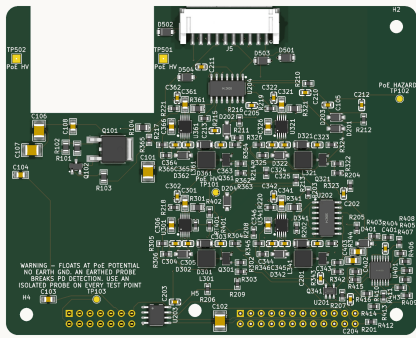
rf-de-20m 4 layers 120 × 80 mm 80 footprints stopped at manufacturability checks

A 20 MHz Class E RF power stage: 200 W into 50 ohms from a 40 V bus. A PWM drive arrives on an SMA input, a GaN gate driver buffers it, one ground-referenced eGaN FET switches, and a resonant tank plus an L-match transforms the 4.6-ohm load line up to 50 ohms at the output SMA. The topology and the two core parts were fixed by the owner after a costed trade study; hwde did the rest.



lumina-par 4 layers 100 × 80 mm 158 footprints stopped at verification

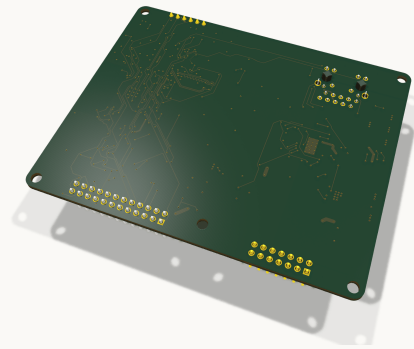
The RGBW PAR daughter board that stacks on the carrier. It is bound by the carrier's frozen connector document — a hard input it may never redefine — and by a re-derived power budget of 8.6 to 20 W depending on which power-over-Ethernet class the fixture gets. The densest board here at 158 footprints, and the only one whose outline is not a rectangle. The silkscreen warns that the board floats at the power-over-Ethernet potential, so an earthed probe on it breaks the fixture's power negotiation.



lumina-par, straight down

top-down view

So the routing shows. Four layers, 158 footprints, laid out and routed to the same rules as everything else.

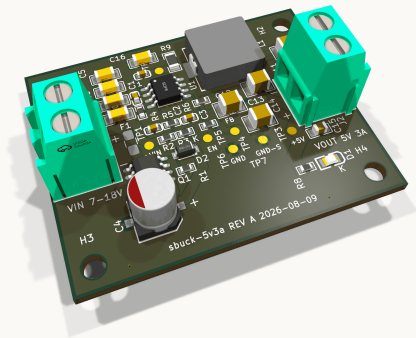


lumina-carrier, underneath

bottom view

Every board here is built with parts on one side only, so the back carries routing and vias and nothing else. This is the fab's view of it.

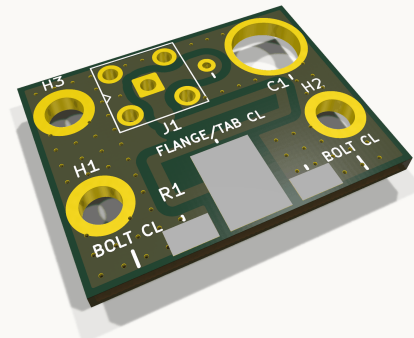
Power



sbuck-5v3a

4 layers 50 × 40 mm 44 footprints package ready

7–18 V in, a regulated 5.0 V $\pm 2\%$ out at 3 A, synchronous buck. An open-frame power module.

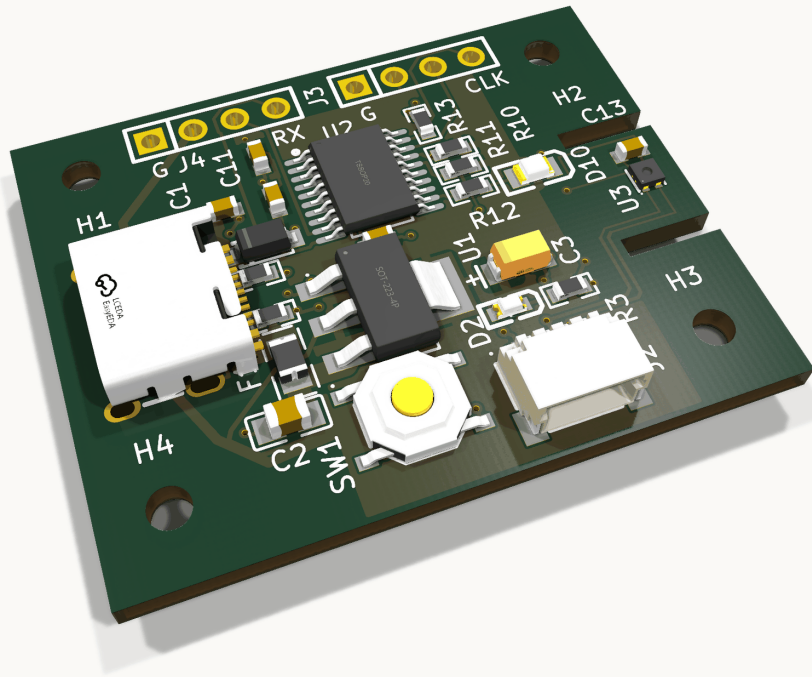


rf-term-150w

2 layers 26 × 20 mm 6 footprints package ready

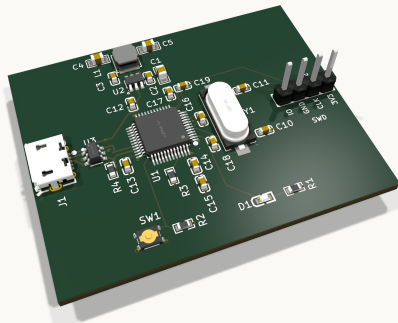
A 50-ohm, 150 W dummy load head, DC to 25 MHz. The element bolts to a heatsink the user supplies; the board is only the RF launch and the mechanical interface. Shown close to bare: one of its six footprints has a 3D model in the repository, so the connector and the element are not drawn.

Microcontroller boards



g0-sense 2 layers 35.8 × 28.3 mm 30 footprints package ready, release attestation built

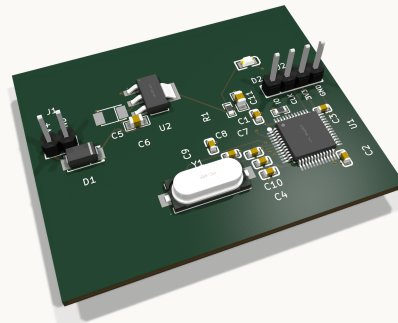
A USB-C powered temperature and humidity node. USB-C supplies 5 V, an LDO makes 3.3 V, an STM32G030 running on its internal oscillator reads a Sensirion SHT4x over I²C. The same bus comes out on a Qwiic connector; readings go out over a UART header; programming is over SWD. Designed end-to-end in one unattended container run.



usb-buck

4 layers 50 × 40 mm 28 footprints package ready

An STM32F103 USB full-speed device board. Micro-B USB carries both power and data; LED, button and an SWD header.



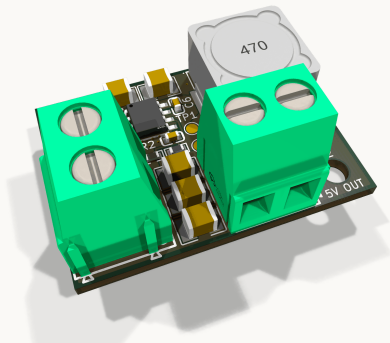
stm32-blinky

2 layers 50 × 40 mm 20 footprints package ready

The simplest useful board: STM32F103, an 8 MHz crystal, one LED, an LDO and an SWD header.

Building blocks

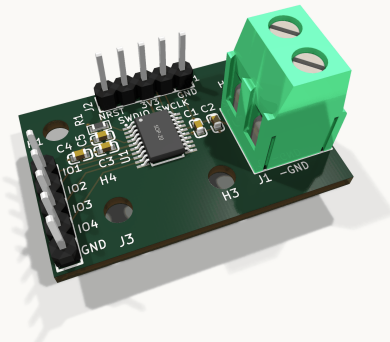
Five deliberately bare boards, each one block and nothing else. They exist to be studied and measured on a bench, so protection, filtering, indicators and spare rails are left out on purpose.



bb-buck

2 layers 35 × 25 mm 20 footprints

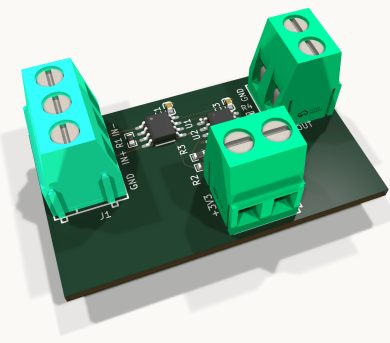
18–30 V in, 5 V at 2 A out on a screw terminal. The converter and exactly what its datasheet requires.



bb-mcu

2 layers 34.8 × 22.3 mm 14 footprints

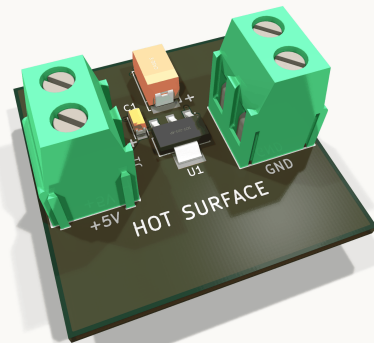
One MCU, a 3.3 V rail in, its debug header and four GPIO. No regulation, no second rail.



bb-amp

2 layers 48 × 28.3 mm 14 footprints

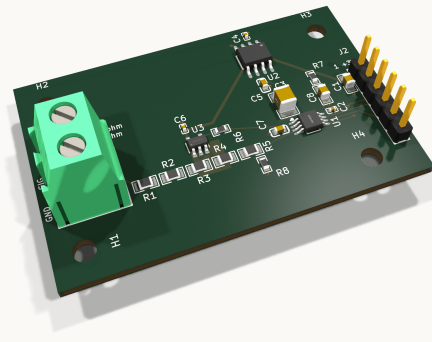
A 0–20 mV differential bridge signal in, 0–3.3 V single-ended out, DC to 1 kHz. Nominally 165 V/V.



bb-ldo

2 layers 34.7 × 34.7 mm 5 footprints

5 V in, 3.3 V at 500 mA out, holding regulation in still air on the board's own copper.



bb-adc

2 layers 54.8 × 34.9 mm 25 footprints

0–5 V in on a screw terminal, 12 bits or better at up to 10 kSa/s out over a 0.1-inch header. DC accuracy over speed.

Two more boards — a 5V/3A buck and a strobe daughter board — stopped before layout and have no PCB to show. They are in the repository at the stage they reached.

What it can and cannot do yet

This is a supervised engineering assistant, not an unattended release system. Boards have been designed, fabricated and ordered with it, and the checker corpus is real. It is still a system a human engineer drives and signs off.

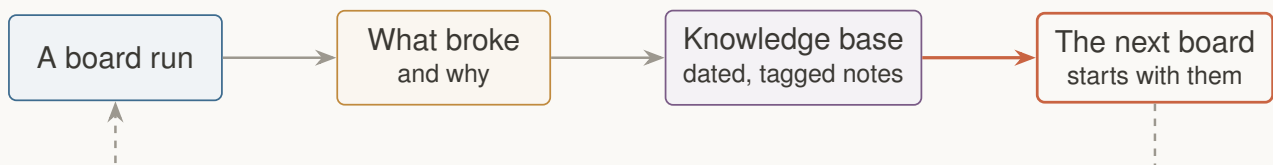
What holds

- Fourteen boards from written briefs to routed layouts, including a 4-layer 200 W RF stage and a novel 100 W USB-C PD design.
- Two ordered from a factory and built, their order numbers tracked to shipped in their own files.
- Every check is a script with the same contract, and its findings are recorded per check rather than as a single verdict.
- Everything a run produced is a file in that board's folder, readable without hwde.

What does not

- A green gate proves the checks that *ran*, not the checks it lists. Read the per-check report, not the verdict.
- The pipeline stage a board reached is not a release certificate. Today a person decides what is releasable.
- There is no public manufacturability API at the factory, so that review stays a human step in a browser.
- Spending money is deliberately hard. One code path can place an order. It is 2-layer only, it refuses a board that already records an order, and it will not run without a fresh quote, a matching design hash and a confirmation a person types out naming the board, the quantity and the total.
- The electrical and thermal checks are engineering screens with stated accuracy. They are not certification, and nothing here is safety-certified for mains, medical, automotive or aerospace use.

How it gets better



Every run writes down what broke and why, dated and tagged, and that goes into the knowledge base the next run reads first. The rule is that a gotcha is recorded the moment it is hit, so the next board does not pay for it again. It is the same reason the boards above get harder as they go.

Renders produced from the board files in this repository with KiCad's command-line renderer. Board dimensions, layer counts and footprint counts read from those same files; the stage each board reached read from its recorded state.